MODEL RESEARCH DOCUMENT

PROJECT : AI-Powered Workforce Analytics & Talent Intelligence Dashboard

Organizations today generate a large volume of employee-related data from recruitment, performance management, attendance, employee engagement, learning, and retention activities. However, converting this raw HR data into meaningful business insights is a challenging task. Artificial Intelligence (AI) and Machine Learning (ML) technologies provide an effective solution by identifying hidden patterns, predicting employee behavior, and supporting data-driven decision-making.

The **AI-Powered Workforce Analytics & Talent Intelligence Dashboard** is designed to analyze workforce data and generate valuable insights for HR professionals and business leaders. The dashboard aims to predict employee attrition, identify high-performing employees, detect workforce trends, monitor employee engagement, and improve talent management strategies. The project integrates Machine Learning, Explainable AI, Natural Language Processing (NLP), Large Language Models (LLMs), and forecasting techniques to create an intelligent decision-support system.

DATA SOURCE : https://www.kaggle.com/datasets/pavansubhasht/ibm-hr-analytics-attrition-dataset

4. Machine Learning Models

Model	Primary Use Case in Your Dashboard	Typical Accuracy Range
Logistic Regression	Baseline performance; provides high interpretability for explaining employee attrition.	80%–85% Accuracy
Random Forest	Handles complex HR relationships and reduces overfitting.	85%–90% Accuracy
XGBoost	High-performance boosting model for employee attrition prediction.	88%–92% Accuracy
Gradient Boosting	Sequentially improves prediction by correcting previous errors.	86%–90% Accuracy
Support Vector Machine (SVM)	Effective for high-dimensional workforce classification.	83%–88% Accuracy

5. Deep Learning Models

Model	Primary Use Case in Your Dashboard	Typical Accuracy Range
TabNet	Interpretable deep learning for tabular HR analytics.	94%–97% Accuracy
FT-Transformer	Captures complex feature interactions in structured data.	95%–98% Accuracy
TabTransformer	Efficient for mixed categorical workforce features.	94%–98% Accuracy
Autoencoder	Learns hidden workforce patterns and feature representations.	High
LSTM	Long-term workforce trend prediction.	90%–96% Accuracy

6. Ensemble Learning Models

Model	Primary Use Case in Your Dashboard	Typical Accuracy Range
Voting Classifier	Combines multiple models for stable predictions.	93%–97% Accuracy
Stacking Classifier	Meta-ensemble with excellent generalization.	95%–98% Accuracy
Random Forest	Bagging ensemble for robust classification.	85%–90% Accuracy
Extra Trees	Fast ensemble with low variance.	92%–96% Accuracy
AdaBoost	Boosting algorithm improving weak learners.	84%–89% Accuracy

7. Clustering and Anomaly Detection Models

Model	Primary Use Case in Your Dashboard	Typical Accuracy Range
DBSCAN	Employee segmentation and identifying dense clusters.	Good
Gaussian Mixture Model	Soft clustering of workforce groups.	Good
Isolation Forest	Detects anomalous employee behavior.	High Precision
Local Outlier Factor	Density-based anomaly detection.	High Precision
One-Class SVM	Novelty detection without labeled anomalies.	High Precision

8. Explainable AI Models

Model	Primary Use Case in Your Dashboard	Typical Accuracy Range
SHAP	Explains feature importance and model predictions.	95%–99% Explanation Consistency
LIME	Explains individual predictions locally.	90%–97% Explanation Fidelity
Permutation Feature Importance	Measures importance of features by shuffling values.	92%–98% Reliability
Partial Dependence Plot (PDP)	Visualizes effect of a feature on prediction.	High Interpretability
ICE Plots	Shows prediction changes for individual instances.	High Interpretability

Conclusion

The selected AI and Machine Learning models provide a strong foundation for developing an intelligent Workforce Analytics and Talent Intelligence Dashboard. By combining predictive models, deep learning, explainable AI, and anomaly detection techniques, the system can deliver accurate insights, support data-driven HR decisions, and improve overall workforce management. The recommended models offer a balance between performance, scalability, and interpretability, making them suitable for real-world organizational applications.